

Semester Project for Soft Computing Course

(Total Seven projects* are in this list along with the experimental datasets)

(Note: Some project from the Image Processing course will carry forward based on the size of the project and outcome shown by group in last semester)

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Evaluation pattern: Progress of the project will be evaluated after ever 15 days in the Machine Vision and Intelligence lab.

Expected Outcome: Project Report (In form of research article), Python code with ppts.

Marks: Total 20 marks (evaluation will be continuous process)

Set - 1

Epileptic Spike Detection in EEG Signal Using Graph Neural Networks

Related Works:

- Graph neural networks in EEG spike detection

https://www.sciencedirect.com/science/article/pii/S093336572300177X?casa_token=1nzjMQZBiW0AAAAA:mUAZvLPq_k15MrTNSmel_DWqEUr2Edb4eJc4XhgnXKVr9LLQwCW6qmmy5dIxs1dC0mhqPtoUsHR

- A Sequential Graph Convolutional Network with Frequency-domain Complex Network of EEG Signals for Epilepsy Detection

https://ieeexplore.ieee.org/abstract/document/9313232?casa_token=1Wq_ZAdjnYAAAAAA:mZOmg1Rlu2lqZSpH-c9XE2se9QvgQK6muqrbgNWDIoweOyp7C9mOAFk96Alcx2fgeMdzPbDp2bWsFg

- Graph-generative neural network for EEG-based epileptic seizure detection via discovery of dynamic brain functional connectivity

[Graph-generative neural network for EEG-based epileptic seizure detection via discovery of dynamic brain functional connectivity | Scientific Reports](#)

Example Datasets: Available in the paper itself.

Deep Fake Detection

Related Works :

- Analysis Survey on Deepfake Detection and Recognition with Convolutional Neural Networks

https://ieeexplore.ieee.org/abstract/document/9799858?casa_token=Br-0AVMdv4MAAAAA:pSXRg73H0DiMSLaRNWnfPNPzPk7xMo_wBQnGAQiOFc_p8IR7fggRJjrdkXCy2K2drGNdLW9Pogw97A

- M2TR: Multi-modal Multi-scale Transformers for Deepfake Detection

https://dl.acm.org/doi/abs/10.1145/3512527.3531415?casa_token=IsCBhYMvIBsAAAA:UlgV0REtaEr0NOoM4kWNViAQvmDxR_lvOEgCkf46aTgpJM3Sb3rT-shMQtmS6WHiWC3aDV393XljAQ

- Deepfake Video Detection Using Convolutional Vision Transformer

[\[2102.11126\] Deepfake Video Detection Using Convolutional Vision Transformer](#)

- ADT: Anti-Deepfake Transformer

https://ieeexplore.ieee.org/abstract/document/9746888?casa_token=SEQLSu-EcPQAAAAA:9p6W38nR2Q6Kdr6wUIGk06XJ5yxfY9SxLdPsMYRSbgW5YG-Fs5_RNDLQrjiYyXkeLTdwJpeMSXu-6w

Example Dataset :

- [\[2006.07397\] The DeepFake Detection Challenge \(DFDC\) Dataset](#)

GAN in Signature Classification

Related Works :

- OffSig-SinGAN: A Deep Learning-Based Image Augmentation Model for Offline Signature Verification

https://cdn.techscience.cn/files/cmc/2023/TSP_CMC-76-1/TSP_CMC_35063/TSP_CMC_35063.pdf

- Cluster GAN-based model for Signature Generation and Verification
https://ieeexplore.ieee.org/abstract/document/10273980?casa_token=eFgoXu4dsocAAAAA:9qzz-zEN1PcBx6i993tWqX2di7B17sdZ4n-z4gwHq40cXutFRWKzODzXs_eDvu1PaqF739JWZd8hzw

- Deep learning-based data augmentation method and signature verification system for offline handwritten signature

<https://link.springer.com/article/10.1007/s10044-020-00912-6>

- Signature handwriting identification based on generative adversarial networks

<https://iopscience.iop.org/article/10.1088/1742-6596/1187/4/042047/meta>

Example Dataset :

- CEDAR
- GPDS
- MCYT

GAN in Image Classification

Related Works :

- Generative Adversarial Networks for Hyperspectral Image Classification

https://ieeexplore.ieee.org/abstract/document/8307247?casa_token=2EQgA_1_e_nUAAAAA:8ISpakDd1txCDheRVQHRLyBQFmhSDM1V1yfzyrqNd5tRWfYXOLG0JQ2Ki_sISNKFUSSTF47MLtqSvg

- A GAN-based image synthesis method for skin lesion classification

https://www.sciencedirect.com/science/article/pii/S0169260720302418?casa_token=0FKe5rIZrRQAAAAA:ZI1AdfolgcMDyXk3RoNFP54_ntbRd6RHlkPe3NssSnE_sxWj1IGXXBHJliJOQligCBiptkzv6jPn2

- GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification

https://www.sciencedirect.com/science/article/pii/S0925231218310749?casa_token=YJFtyheKrHAAAAAA:-IWvAPEagEF8cwnBK6ahxR-geTj4IjbNGnaGcuffdGIE0SXujm0_PBLz60dyunwNUKsg7SnBBRln

- Image Classification and Generation Based on GAN Model

https://ieeexplore.ieee.org/abstract/document/9731058?casa_token=E4fGICHITGIAAAAA:OA7zZNPBwNp80uBfHW-_VoGtKN_Zmk4qFXqAPhD1XAmwBUHUXuLeF0qYoK_znvW3cWUbnOEYg9q9Sw

Example Dataset :

Multi-Scale CNN for EEG BCI Motor Imagery Classification

Related Works :

- A Multi-Branch 3D Convolutional Neural Network for EEG-Based Motor Imagery Classification

<https://ieeexplore.ieee.org/document/8820089>

- A novel multi-branch hybrid neural network for motor imagery EEG signal classification

https://www.sciencedirect.com/science/article/pii/S1746809422002403?casa_token=UoDdUILe0UcAAAAA:J6EUMohDW2VuadxomVA2OPqGVmDgVe_Mov0JIOy4sRDstjoGZj-ZCtgXMgEdfiUFq-eiEDz-jf1n

- Motor Imagery EEG Classification Based on Transfer Learning and Multi-Scale Convolution Network

<https://www.mdpi.com/2072-666X/13/6/927>

- MMCNN: A Multi-branch Multi-scale Convolutional Neural Network for Motor Imagery Classification

https://link.springer.com/chapter/10.1007/978-3-030-67664-3_44

Example Dataset :

- BCI Competition IV dataset 2a and HGD

Localisation of Epileptic Foci Based on Simultaneous EEG–fMRI Data

Related Works :

- Component-related BOLD response to localise epileptic focus using simultaneous EEG-fMRI recordings at 3T

<https://www.sciencedirect.com/science/article/abs/pii/S0165027019301219>

- Localisation of Epileptic Foci Based on Simultaneous EEG–fMRI Data

<https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2021.645594/full>

- Integration of fMRI and simultaneous EEG: towards a comprehensive understanding of localisation and time-course of brain activity in target detection

<https://www.sciencedirect.com/science/article/abs/pii/S1053811904000084>

- Localising Epileptic Foci Using Simultaneous EEG-fMRI Recording: Template Component Cross-Correlation

<https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2021.695997/full>

Example Dataset :

UNFOLD: 3-D U-Net, 3-D CNN, and 3-D Transformer-Based Hyperspectral Image Denoising

<https://ieeexplore.ieee.org/abstract/document/10302305>

Abstract :

Hyperspectral image (HSI) contains information at several spectrums, thus, extensively used in several real-world domains, including remote sensing [1], classification [2], [3], [4], agriculture [5], and marine monitoring [6]. It is represented as a 3-D array, incorporating two spatial and one spectral dimension. Unfortunately, noise can be added during the HSI sensing due to various factors, including limited light, photon effects, and atmospheric interference [7], thereby degrading HSI quality. This issue is mitigated by HSI denoising.

Example Dataset :

ICVL hyperspectral dataset <https://paperswithcode.com/dataset/icvl-hsi>

Note: Use institute LAN to download base papers.